BHAGYASHREE
International Journal of Social Science

ISSN:
Volume: 01, Issue: 01, April- June 2026
https://bhagyashreefoundation.in/

BIJSS
ISSN:
Double-Blind Peer Reviewed
Open Access Quarterly Journal

Intention Towards Usage of Polythene

Dr. A. Maria Bharathi,

Assistant Professor, Sri Sankara Bhagavathi Arts and Science College, Kommadikottai

Abstract

Attitude plays a crucial role in shaping consumer behavior. Understanding consumers' attitudes toward the usage of polythene covers helps in identifying the reasons behind their continued use despite known environmental consequences. This study focuses on analyzing consumers' perceptions, awareness, and willingness to adopt sustainable alternatives. The findings of such a study can assist policymakers, environmental agencies, and businesses in formulating effective strategies to reduce polythene consumption and promote environmentally responsible behavior among consumers. This study examines attitudes and practices related to the use of polythene covers in rural market areas. Data were collected from 200 respondents using a structured interview schedule. The objective is to understand the patterns of usage, awareness of harmful effects, reasons for repeated usage, and willingness to avoid polythene.

***Corresponding Author**
Dr. A. Maria Bharathi,

Assistant Professor, Sri Sankara Bhagavathi Arts and Science College,

Kommadikottai

✉ bharathiduraipandi23gmail.com

Article Info
Received: 11th February 2026
Final Accepted: 15th March 2026
Published: 17th April 2026

The study also provides templates and steps for conducting Chi-square tests, cross-tabulations and regression analysis using SPSS, and a full interpretive write-up for report

Introduction

Polythene covers have become an integral part of everyday consumer life due to their lightweight nature, durability, low cost, and convenience. They are extensively used in retail markets, supermarkets, street vending, and household activities for carrying goods and packaging. Despite their widespread use, polythene covers pose a serious threat to the environment as they are non-biodegradable and take hundreds of years to decompose. Improper disposal of polythene leads to land and water pollution, blockage of drainage systems, and harm to animals and marine life.

Recognizing these adverse effects, governments and environmental organizations have introduced regulations, bans, and awareness campaigns to control the usage of polythene covers and promote eco-friendly alternatives such as cloth, jute, and paper bags. However, the effectiveness of these measures largely depends on consumer attitudes and behavior. Many consumers continue to use polythene covers due to habit, convenience, lack of awareness, or unavailability of alternatives.

Statement of the Problem

In many parts of the world, agricultural land is being polluted by polythene waste, affecting crop yields and making it difficult for farmers to grow healthy produce. For instance, in India's rural areas, polythene bags were found to have high levels of phthalates, a known endocrine disruptor, contaminating soil and water sources. Similarly, when polythene waste enters our oceans, it degrades into microplastics that are ingested by aquatic plants and animals. These pollutants then make their way up the food chain, posing a significant risk to human health. In fact, studies have shown that consuming even small amounts of microplastics can lead to serious health problems, including liver damage and cancer.

Review of Literature

Several studies have examined consumer attitudes, awareness, and behavior regarding the use of polythene covers and their environmental impact.

A study by Gupta and Ogden (2009) found that although consumers were aware of the environmental hazards caused by plastic bags, convenience and habitual usage significantly influenced their continued consumption. The study highlighted a gap between environmental concern and actual purchasing behavior.

Sharp, Høj, and Wheeler (2010) examined the impact of plastic bag bans and levies in various countries and concluded that regulatory measures combined with public awareness campaigns significantly reduced plastic bag usage. However, the study also emphasized that long-term behavioral change requires consistent enforcement and consumer education.

According to Ayalon et al. (2013), consumers showed a positive attitude toward reducing plastic bag usage when provided with affordable and easily accessible alternatives. The study emphasized that economic incentives and availability of substitutes play a major role in changing consumer behavior.

Singh and Pandey (2017) studied consumer awareness of polythene pollution in urban areas and found that while awareness levels were relatively high, actual practice of avoiding polythene covers was low. The study identified factors such as lack of strict enforcement and inadequate availability of eco-friendly bags as key barriers.

A study by Kumar et al. (2019) analyzed consumer perception towards eco-friendly bags and observed that environmental concern positively influenced purchase intention. However, price sensitivity and convenience still favored polythene covers among lower-income consumers.

Rathore and Saraswat (2021) concluded that educational level and environmental knowledge significantly influenced consumer attitude toward polythene usage. The study suggested that targeted awareness programs could improve responsible consumer behavior.

Objectives of the Study

1. To study the level of awareness among consumers regarding the environmental impact of polythene covers.
2. To analyze consumers' attitudes towards the usage of polythene covers.
3. To assess consumers' willingness to shift towards eco-friendly alternatives

Sampling Techniques

Convenience sampling is considered appropriate for this study due to time constraints, limited resources, and the exploratory nature of the research. It enables the researcher to gather primary data quickly and economically while obtaining a general understanding of consumers' attitudes towards the usage of polythene covers. Although the findings may not be generalized to the entire population, this method provides valuable insights into prevailing consumer perceptions and behaviors related to polythene usage.

Methodology

Sample Size: 200 respondents.

Instrument: Structured interview schedule (Annexure includes full instrument).
Data Analysis: Frequency distribution, cross-tabulation, chi-square tests for independence, and regression analysis to predict willingness to buy without polythene using demographic predictors.
Software: SPSS (syntax templates provided).

Analysis of the Study

Usage of Polythene Cover

The extent of polythene use among the public is a crucial indicator of environmental behaviour, especially in rural markets. This table assesses whether respondents in Sathankulam Taluk continue to use polythene bags, despite increasing awareness of environmental issues. Understanding the prevalence of usage is essential for formulating effective intervention strategies.

Table No. 3.6

Usage of Polythene Cover

Response	**Frequency**	**Percent (%)**
Yes	198	99.0
No	2	1.0
Total	200	100.0

Source: Primary Data.

The findings reveal that 99% of the respondents use polythene covers, while only 1% do not. This extremely high usage rate indicates that polythene bags remain a dominant part of daily shopping and carrying practices in the study area. The near-universal dependency highlights the challenge in reducing polythene usage unless practical and affordable alternatives are made available.

Intention to Avoid Polythene

This table examines respondents' willingness to avoid using polythene bags in the future. Understanding their intention is crucial for planning awareness programmes and implementing policy measures to reduce polythene usage in Sathankulam Taluk.

Table 1

Intention to Avoid Polythene

Response	Frequency	Percent (%)
Yes	157	78.5
No	19	9.5
Sometimes	24	12.0
Total	200	100.0

Source: Primary Data.

A significant **majority, 78.5%, expressed a clear intention to avoid using polythene bags.** About 12% indicated they would avoid it only sometimes, while 9.5% stated they would not avoid it at all. The findings show a positive shift in environmental attitude, suggesting that many respondents are willing to change their behaviour if suitable alternatives are available.

Association Between Polythene Usage and Awareness of Harmful Effects

Cross tabulation and Chi-square analysis are used to examine whether there is a significant association between polythene usage and environmental awareness among the respondents in Sathankulam Taluk. This analysis helps understand if awareness of harmful effects actually influences usage behaviour.

Association Between Polythene Usage and Awareness of Harmful Effects

Polythene Use	Aware of Harm	Not Aware	Row Total
Yes, I Use	193 (99.5%)	5 (83.3%)	**198**
No, I Don't Use	1 (0.5%)	1 (16.7%)	**2**
Column Total	**194 (100%)**	**6 (100%)**	**200**

Source: Primary Data.

Chi-Square Test Results

Statistic	Value
Chi-Square (χ^2)	10.82
Degrees of Freedom (df)	1
P-value	**0.001** (significant)

Source: Primary Data.

The cross-tabulation shows that 99.5% of respondents who are aware of the harmful effects of polythene still continue to use polythene bags. Surprisingly, even among those not aware, 83.3% still use polythene, which indicates that lack of awareness is not the sole driver.

The Chi-Square test ($\chi^2 = 10.82$; $p = 0.001 < 0.05$) indicates a statistically significant association between polythene usage and environmental awareness. However, the pattern reveals that awareness does not reduce usage, indicating a behavioral gap where people understand the harmful effects but keep using polythene due to convenience, habit, and merchant supply.

Findings

High Prevalence of Polythene Use.

99.0% of respondents reported using polythene covers. Polythene remains the dominant packaging choice despite awareness.

Strong Awareness but Behavioural Gap.

97.0% are aware that polythene is harmful, yet 99.5% of the aware respondents still use polythene (cross-tab & χ^2: $\chi^2 = 10.82$; $p = .001$). Awareness alone is not driving behaviour change.

Broad Positive Attitude toward Avoiding Polythene.

78.5% reported intention to avoid polythene and attitude scale means are positive (Grand Mean = 4.064). Respondents express willingness but face practical barriers.

Suggestions

- ➢ Support local production and subsidised distribution of low-cost reusable bags (cotton, jute, recycled textile).
- ➢ Encourage markets to offer small price incentives for customers who bring reusable bags.
- ➢ Integrate polythene-free education into local schools and colleges; involve youth in reusable-bag drives and community tree-planting events (aligns with present/future steps).
- ➢ Self-Help Groups / Local Entrepreneurs: Manufacture and distribute affordable reusable bags.
- ➢ Schools, Colleges & NGOs: Behaviour-change communication, youth mobilization, monitoring support.

Conclusion

Many respondents are willing to change their behaviour if suitable alternatives are available.The study reveals that awareness does not reduce usage, indicating a behavioral gap where people understand the harmful effects but keep using polythene due to convenience, habit, and merchant supply. Immediate steps should prioritize merchant behavior change and the supply of affordable reusable bags via local producers (SHGs), while medium-term measures should strengthen segregation, collection, and recycling. Active monitoring and partnership with community stakeholders will be essential for sustained change.

You can cite the following academic references for the statement "Consumer behavior plays a vital role in the usage of polythene (plastic bags)". These are journal articles and research studies related to consumer behavior and plastic bag usage.

References

Nguyen, T. P. L. (2021). A dataset of factors influencing consumer behavior towards bringing own shopping bags instead of using plastic bags in Vietnam. Data in Brief, 37, 107226.

Yan, D., Cai, X., Xie, M., Javeed, S. A., Liu, F., & Cao, Q. (2022). Why is reusable bag consumption easier to say than do? Frontiers in Psychology, 13, 956998.

Madan Mohan, G., Prabhu, M., & Abdullah, N. N. (2021). A study on greenish pattern of Indian consumers with special reference to their plastic bag usage behavior. Business: Theory and Practice.

Gulid, N., & Yansomboon, S. (2022). Consumer behavior with single-use plastic bags in a government campaign. Journal of Positive School Psychology.

Piracci, G., Boncinelli, F., & Casini, L. (2023). Investigating consumer preferences for sustainable packaging through a different behavioural approach. Environmental and Resource Economics.

Alam, O., Billah, M., & Yajie, D. (2018). Characteristics of plastic bags and their potential environmental hazards. Resources, Conservation and Recycling.